University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

First Midterm Exam – Version A

October 2, 2018

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(a) You have 80 minutes to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	7	Total
Maximum	5	3	4	4	2	4	2	24
Grade								

1. Consider the following linear system:

$$\begin{cases} y_1 - 2y_5 &= 3y_2 + 3y_3 + 4 \\ y_1 - y_3 &= 3y_2 + 4y_5 + 2 \\ 2y_1 + y_4 - 9 &= 6y_2 + 6y_3 + y_5. \end{cases}$$

(a) [**4 points**] Write down the general solution of this linear system.

Solution: The augmented matrix of the system is $\left[\begin{array}{ccccc|c} 1 & -3 & -3 & 0 & -2 & 4 \\ 1 & -3 & -1 & 0 & -4 & 2 \\ 2 & -6 & -6 & 1 & -1 & 9 \end{array}\right]$

The R.E.F of the the augmented matrix is obtained as follows:

$$\left[\begin{array}{ccccc|c} 1 & -3 & -3 & 0 & -2 & 4 \\ 1 & -3 & -1 & 0 & -4 & 2 \\ 2 & -6 & -6 & 1 & -1 & 9 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccccc|c} 1 & -3 & -3 & 0 & -2 & 4 \\ 0 & 0 & 2 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 & 3 & 1 \end{array}\right] \text{ E.F.}$$

$$\xrightarrow{R_2 \to \frac{1}{2}R_2} \left[\begin{array}{ccccc|c} 1 & -3 & -3 & 0 & -2 & 4 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & 3 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 + 3R_2} \left[\begin{array}{ccccc|c} 1 & -3 & 0 & 0 & -5 & 1 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & 3 & 1 \end{array}\right] \text{ R.E.F.}$$

The linear system corresponding to the last augmented matrix is:

$$\begin{cases} y_1 - 3y_2 - 5y_5 &= 1 \\ y_3 - y_5 &= -1 \\ y_4 + 3y_5 &= 1 \end{cases}$$

From the E.F. or the R.E.F. we find out that y_1, y_3, y_4 are basic, and y_2, y_5 are free. The general solution is:

$$\begin{cases} y_1 = 3y_2 + 5y_5 + 1 \\ y_2 = \text{free} \\ y_3 = y_5 - 1 \\ y_4 = -3y_5 + 1 \\ y_5 = \text{free} \end{cases}$$

(b) [**1 point**] Give an explicit solution of this linear system such that $y_2 = -1$ and $y_5 = 3$.

Solution: From the general solution it follows that

$$y_1 = 13,\ y_3 = 2,\ y_4 = -8.$$

Thus the explicit solution is:

$$y_1 = 13,\ y_2 = -1,\ y_3 = 2,\ y_4 = -8,\ y_5 = 3.$$

2. Set

$$A = \begin{bmatrix} 1 & 0 & -2 & 0 & -3 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}.$$

(a) **[2 points]** Suppose that $\mathbf{b} = \begin{bmatrix} -3 \\ 1 \\ -1 \end{bmatrix}$. Write down the general solution of the linear system $A\mathbf{x} = \mathbf{b}$ in vector parametric form.

Solution:

The augmented matrix of the system is $\left[\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & -3 & -3 \\ 0 & 1 & 3 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{array}\right]$ which is in reduced echelon form. The linear system with this augmented matrix is:

$$\begin{cases} x_1 & -2x_3 & & -3x_5 & = & -3 \\ & x_2 & +3x_3 & +x_5 & = & 1 \\ & & & x_4 \quad -x_5 & = & -1 \end{cases}$$

From the augmented matrix, we know that x_1, x_2, x_4 are basic. Therefore the general solution is

$$\begin{cases} x_1 = 2x_3 + 3x_5 - 3 \\ x_2 = -3x_3 - x_5 + 1 \\ x_3 = \text{free} \\ x_4 = x_5 - 1 \\ x_5 = \text{free} \end{cases}$$

and in vector parametric form:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_3 + 3x_5 - 3 \\ -3x_3 - x_5 + 1 \\ x_3 \\ x_5 - 1 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -3 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

(b) **[1 point]** Does there exist a vector $\mathbf{b} \in \mathbb{R}^3$ for which the linear system $A\mathbf{x} = \mathbf{b}$ is inconsistent? You should justify your answer.

Solution: NO, because the coefficient matrix of the system is always A, and every row in A has a pivot position. Therefore the Guaranteed Existence of Solution criterion is met.

3. [**4 points**] Determine all the values of the parameter a for which the linear system

$$\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & -2 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

is consistent. For each of these values of a, how many solutions does the linear system have? You should justify your answers.

Solution: The augmented matrix of this system can be reduced as follows

$$\left[\begin{array}{ccc|c} 1 & 2 & a & 1 \\ 0 & 1 & -2 & 3 \\ -1 & 1 & 3 & 5 \end{array}\right] \xrightarrow{R_3 \to R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 2 & a & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 3 & a+3 & 6 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 2 & a & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & a+9 & -3 \end{array}\right]$$

If $a + 9 \neq 0$, that is, if $a \neq -9$, then the above matrix will be in E.F. and the first 3 columns will be pivot columns, whereas the rightmost column is not a pivot column. The latter implies that the linear system is consistent, and the former implies that all of the variables become basic, hence in these cases the solution is unique.

If $a + 9 = 0$, that is, if $a = -9$, then again the matrix is in E.F. and this time the rightmost column is now a pivot column (indeed the pivot columns are now the 1st, 2nd, and 4th). Therefore the system will be inconsistent.

4. [**4 points**] For each of the following statements, indicate if it is true or false. You will receive 1 point for every correct answer and lose .5 points for every incorrect answer (thus, if you leave the answer space blank, you will not receive negative points). You will not receive a negative total score on this question.

_____ If the rightmost column of the coefficient matrix of a linear system contains a pivot position, then the linear system must be inconsistent.

Solution: False.

_____ A homogeneous linear system with 4 equations and 5 variables always has infinitely many solutions.

Solution: True.

_____ For any two given vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, and any two given scalars $c_1, c_2 \in \mathbb{R}$, we always have $c_1\mathbf{a} + c_2\mathbf{b} \in \text{Span}\{\mathbf{a}, \mathbf{b}\}$.

Solution: True.

_____ Two row-equivalent matrices which are both in echelon form must be equal.

Solution: False

5. Compute the following:

(a) **[1 point]** $c_1 \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix} - c_3 \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$ for $c_1 = -2$, $c_2 = 1$, and $c_3 = 3$.

Solution: $\begin{bmatrix} -6 \\ -8 \\ 11 \end{bmatrix}$

(b) **[1 point]** $\begin{bmatrix} -2 & 1 & -\frac{1}{2} \\ 3 & 1 & 4 \\ -2 & -5 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$.

Solution: $\begin{bmatrix} -2 \\ -6 \\ 3 \end{bmatrix}$

6. [**4 points**] Set

$$\mathbf{a}_1 = \begin{bmatrix} 3 \\ 0 \\ 0 \\ -3 \end{bmatrix}, \ \mathbf{a}_2 = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 3 \end{bmatrix}, \text{ and } \mathbf{a}_3 = \begin{bmatrix} 4 \\ 1 \\ -3 \\ -2 \end{bmatrix}.$$

Does the vector $\mathbf{b} = \begin{bmatrix} -1 \\ 2 \\ -7 \\ 2 \end{bmatrix}$ belong to $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$? You should justify your answer.

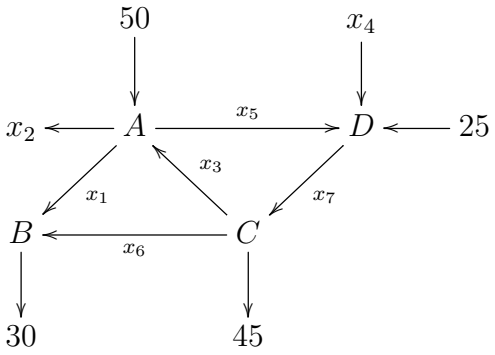
Solution: We should reduce the following matrix to an E.F.:

$$\begin{bmatrix} 3 & -1 & 4 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & -1 & -3 & -7 \\ -3 & 3 & -2 & 2 \end{bmatrix} \xrightarrow{R_4 \to R_4 + R_1} \begin{bmatrix} 3 & -1 & 4 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & -1 & -3 & -7 \\ 0 & 2 & 2 & 1 \end{bmatrix}$$

$$\xrightarrow[R_4 \to R_4 - 2R_2]{R_3 \to R_3 + R_2} \begin{bmatrix} 3 & -1 & 4 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -2 & -5 \\ 0 & 0 & 0 & -3 \end{bmatrix} \text{ E.F.}$$

Since the rightmost column of the E.F. is a pivot column, it follows that $\mathbf{b}$ is <u>not</u> a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$.

7. [**2 points**] Consider the traffic flow described by the following diagram. The letters A through D label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 7$. (You do not need to solve the linear system.)

Solution:

$$\begin{cases} A: & 50 + x_3 = x_1 + x_2 + x_5 \\ B: & x_1 + x_6 = 30 \\ C: & x_7 = x_3 + x_6 + 45 \\ D: & 25 + x_4 + x_5 = x_7 \\ \text{total}: & 25 + 50 + x_4 = x_2 + 30 + 45 \end{cases}$$